WORKED SOLUTION

STEP III 2014, Question 13

Probability and Statistics • Probability Generating Functions

STEP III 2014, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q13

The key idea

Condition on what happens in the first round. The game has no memory: after any round in which it carries on, what follows is a brand-new copy of the same game, so the *extra* score from that point on has the same distribution as N itself. That gives an equation for G in terms of itself, which we solve and expand as a geometric series.

Solution

Write $p_n = P(N = n)$, so $G(t) = \sum_{n\geq 0} p_n t^n$. Let E_1, E_2, E_3 be the events that the first round has outcome 1, 2 or 3, so that $P(E_1) = a$, $P(E_2) = b$ and $P(E_3) = c$.

Part (i)

Given E_1, the game stops with score 0, so $P(N = 0 \mid E_1) = 1$ and $P(N = n \mid E_1) = 0$ for $n \geq 1$. The conditional generating function is therefore

$$\sum_{n\geq 0} P(N = n \mid E_1)\, t^n = 1 \cdot t^0 = 1.$$

Part (ii)

Given E_2, the score is still 0 after round one and the game continues. Since the rounds have the same probabilities and do not depend on what came before, the rounds from round two onwards form a fresh game starting from score 0. Let N' be the score that fresh game produces; it has the same distribution as N. The final score is $0 + N' = N'$, so

$$P(N = n \mid E_2) = P(N' = n) = p_n \quad \text{for every } n \geq 0,$$

and the conditional generating function is $\sum p_n t^n = G(t)$.

Part (iii)

In the same way, given E_3 the score is 1 after round one and the rest is a fresh game with score N', so the final score is $1 + N'$. Hence $P(N = 0 \mid E_3) = 0$ and $P(N = n \mid E_3) = p_{n-1}$ for $n \geq 1$; the conditional generating function is $\sum_{n\geq 1} p_{n-1}t^n = tG(t)$.

By the law of total probability,

$$p_n = a\,P(N = n \mid E_1) + b\,P(N = n \mid E_2) + c\,P(N = n \mid E_3).$$

The coefficient of t^n in $G(t)$ is p_n. On the right, the coefficients of t^n in 1, $G(t)$ and $tG(t)$ are exactly $P(N = n \mid E_1)$, $P(N = n \mid E_2)$ and $P(N = n \mid E_3)$ from parts (i)–(iii). So the coefficients of t^n in $G(t)$ and in $a + bG(t) + ctG(t)$ agree for every $n \geq 0$, which proves

$$G(t) = a + bG(t) + ctG(t).$$

Ask yourself Can we divide by $1 - b - ct$, and is it legitimate to expand as a geometric series?

Rearranging, $G(t)(1 - b - ct) = a$. Since $1 - b = a + c > c > 0$, we have $1 - b - ct \neq 0$ whenever $|t| < (1 - b)/c$, and this range includes $|t| \leq 1$ because $(1 - b)/c = 1 + a/c > 1$. For such t,

$$G(t) = \frac{a}{1 - b - ct} = \frac{a}{1 - b} \cdot \frac{1}{1 - \dfrac{ct}{1 - b}} = \frac{a}{1 - b} \sum_{n\geq 0} \left(\frac{c}{1 - b}\right)^n t^n,$$

since $|ct/(1 - b)| < 1$. Reading off the coefficient of t^n,

$$P(N = n) = \frac{ac^n}{(1 - b)^{n+1}}, \qquad n \geq 0.$$

(Equivalently, the coefficient comparison gives $p_0 = a + bp_0$ and $p_n = bp_n + cp_{n-1}$ for $n \geq 1$, i.e. $p_0 = a/(1 - b)$ and $p_n = \frac{c}{1-b} p_{n-1}$, which leads to the same formula.)

Part (iv)

The mean is $\mu = G'(1)$. Differentiating $G(t) = a(1 - b - ct)^{-1}$,

$$G'(t) = \frac{ac}{(1 - b - ct)^2}, \qquad \text{so} \qquad \mu = G'(1) = \frac{ac}{(1 - b - c)^2} = \frac{ac}{a^2} = \frac{c}{a},$$

using $1 - b - c = a$. Then $1 + \mu = \dfrac{a + c}{a} = \dfrac{1 - b}{a}$, and so

$$\frac{\mu^n}{(1 + \mu)^{n+1}} = \frac{c^n}{a^n} \cdot \frac{a^{n+1}}{(1 - b)^{n+1}} = \frac{ac^n}{(1 - b)^{n+1}} = P(N = n),$$

as required.

Checks

- The probabilities sum to 1: $\sum_n p_n = G(1) = a/(1-b-c) = a/a = 1$.
- A direct argument: ignore the "wasted" outcome-2 rounds. Each remaining round is a stop with probability $a/(a+c)$ or a point with probability $c/(a+c)$, so N is the number of points before the first stop, and $P(N=n) = \left(\frac{c}{a+c}\right)^n \frac{a}{a+c}$, which is the same formula since $a + c = 1 - b$.
- With $a = \frac{1}{5}$, $b = \frac{3}{10}$, $c = \frac{1}{2}$ we get $\mu = \frac{5}{2}$ and $P(N=0) = \frac{2}{7} \approx 0.2857$, $P(N=1) = \frac{10}{49} \approx 0.2041$. A computer simulation of 2×10^5 games gave 0.2855, 0.2038 and a mean score of 2.498.

Common slips

- Writing the conditional generating function in case 3 as $G(t)$ instead of $tG(t)$: the point already scored shifts every power of t up by one.
- Forgetting to say *why* the game after round one is a copy of the original: it relies on each round having the same probabilities, independently of the past.

Beyond STEP

Conditioning on the first step is the basic tool of Markov chain theory, where it is called *first-step analysis*, and the answer here is a geometric distribution in disguise. Part (iv) shows that a geometric distribution on $\{0, 1, 2, \dots\}$ is determined by its mean alone. The same equation-for-G trick handles branching processes, where it leads to the famous result about the probability that a family name dies out. Grimmett and Welsh, *Probability: An Introduction*, treats generating functions and branching processes very readably.

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